

Torrent Regression – Robust Learning Under Corrupted Data

This project implements Torrent Regression and evaluates its robustness under joint corruption (affecting both features and targets). It compares Torrent-FC and Torrent-HYB variants on the Life Expectancy dataset.

File Structure

- feature_selection.ipynb – Step 1: Select top features
- dual_corruption.ipynb – Step 2: Compare FC vs HYB under dual corruption
- torrent_eval.ipynb – Step 3: Evaluate on test data across corruption levels
- Life Expectancy Data.csv – Raw dataset
- Life_Expectancy_Selected_Features.csv – Cleaned + reduced dataset
- Torrent_regression_report.pdf – Final project report
- README.docx – This file

How to Run

Step 1: Feature Selection

Open feature_selection.ipynb

- Loads the original dataset (Life Expectancy Data.csv)
- Cleans missing values
- Applies correlation-based selection
- Saves the result to Life_Expectancy_Selected_Features.csv

Step 2: FC vs HYB Benchmark (Dual Corruption Analysis)

Open dual_corruption.ipynb

- Loads the selected features
- Applies joint corruption to training data
- Compares Torrent-FC and Torrent-HYB
- Outputs plots for: Recovery Error, R^2 , Active Set Stability
- Use this to choose the better method for your data

Step 3: Final Testing (Test Set Evaluation)

Open `torrent_eval.ipynb`

- Uses the best variant (typically Torrent-HYB)
- Trains on corrupted data
- Evaluates on a clean test set
- Plots weight recovery, R^2 scores, convergence

Output

- Recovery error vs corruption plots
- Test performance comparisons
- Active set stability over iterations
- Weight vector visualization (vs ground truth and Lasso)

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